(4) Determine the values of a and b so that the following matrices are singular:

(i) $A=\begin{bmatrix} 7 & 3 \\ -2 & a \end{bmatrix}$ (ii) $B=\begin{bmatrix} b-1 & 2 & 3 \\ 3 & 1 & 2 \\ 1 & -2 & 4 \end{bmatrix}$

(5) If $\cos 2\theta = 0$, determine $\begin{vmatrix} 0 & \cos\theta & \sin\theta \\ \cos\theta & \sin\theta & 0 \\ \sin\theta & 0 & \cos\theta \end{vmatrix}^2$.

(6) Find the value of the product ; $\begin{vmatrix} \log_3 64 & \log_4 3 \\ \log_3 8 & \log_4 9 \end{vmatrix} \times \begin{vmatrix} \log_2 3 & \log_8 3 \\ \log_3 4 & \log_3 4 \end{vmatrix}$.

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EXERCISE 7.5

Choose the correct or the most suitable answer from the given four alternatives.

(1) If $a_{ij}=\frac{1}{2}(3i-2j)$ and $A=[a_{ij}]_{2\times 2}$ is

(1) $\begin{bmatrix} \frac{1}{2} & 2 \\ -\frac{1}{2} & 1 \end{bmatrix}$ (2) $\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ 2 & 1 \end{bmatrix}$ (3) $\begin{bmatrix} 2 & 2 \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$ (4) $\begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ 1 & 2 \end{bmatrix}$

(2) If $2X+\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}=\begin{bmatrix} 3 & 8 \\ 7 & 2 \end{bmatrix}$ then matrix X is

(1) $\begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}$ (2) $\begin{bmatrix} 1 & -3 \\ 2 & -1 \end{bmatrix}$ (3) $\begin{bmatrix} 2 & 6 \\ 4 & -2 \end{bmatrix}$ (4) $\begin{bmatrix} 2 & -6 \\ 4 & -2 \end{bmatrix}$

(3) Which one of the following is not true about the matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{bmatrix}$?

(1) a scalar matrix (2) a diagonal matrix
(3) an upper triangular matrix (4) a lower triangular matrix

(4) If A and B are two matrices such that $A + B$ and AB are both defined, then
(1) A and B are two matrices not necessarily of same order
(2) A and B are square matrices of same order
(3) Number of columns of A is equal to the number of rows of B
(4) $A = B$.

(5) If $A=\begin{bmatrix} \lambda & 1 \\ -1 & -\lambda \end{bmatrix}$, then for what value of $\lambda, A^2 = O$?

(1) 0 (2) ± 1 (3) -1 (4) 1

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